UNIT - IV

TQM TOOLS

Benchmarking - QFD - Taguchi - TPM - FMEA - 7 QC Tools - Process Capability - Six Sigma - New 7 Tools

1. Benchmarking

Benchmarking is a systematic and continuous process of measuring and comparing an organisation's products, services and processes against those of the **best-in-class** organisations, to get information that helps to improve performance. (Camp: "the search for industry best practices that lead to superior performance")

- Started at **Xerox** (1979) against Japanese copier makers.
- Benchmark = a reference point / standard of excellence. Benchmark is not copying - it is **learning and adapting.**

Reasons to benchmark

- Stay competitive; set realistic and credible goals based on facts.
- Meet customer requirements, find out what customers expect.
- Break the "not invented here" syndrome - learn from others.
- Find the gaps in the performance and the best practices; quick improvement and fewer mistakes.
- Helps to be proactive, to change the culture, to understand own process better.
- Provides a basis for **continuous improvement** and strategic planning.

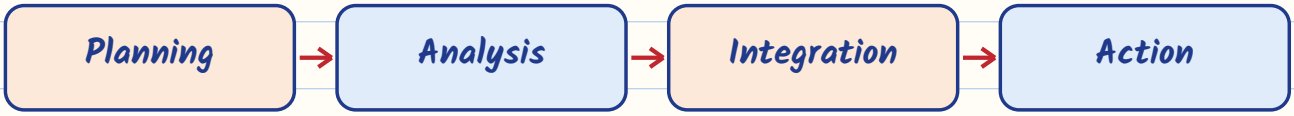
Types of benchmarking

- **Internal** (between units of the same company); **Competitive** (direct competitor); **Functional/Industry** (similar function in different industry); **Generic** (best process irrespective of industry - e.g. Xerox

vs L.L.Bean warehousing).

- Also: performance, process and strategic benchmarking.

Benchmarking process (Xerox 10 steps, grouped into 4 phases)



1. Decide **what** to benchmark (critical success factor).
2. Identify the best competitors / best practices.
3. Identify the **data collection** method and collect the data (surveys, visits, journals, internet).
4. Determine the **current performance gap.**
5. Project future performance levels.
6. **Communicate** the findings and gain acceptance.
7. Establish functional **goals.**
8. Develop **action plans.**
9. **Implement** the plan and monitor the results.
10. Recalibrate benchmarks - continue the cycle.

- Simplified (Besterfield): Plan - Collect - Analyse - Adapt/Improve.

Pitfalls

- Wrong choice of subject; copying blindly; poor partner selection; no top management support; ignoring ethics (a code of conduct: legality, exchange, confidentiality).

Benefits

- Improves quality and productivity; competitive edge; cost reduction; improves customer satisfaction; creates awareness of the best practice.

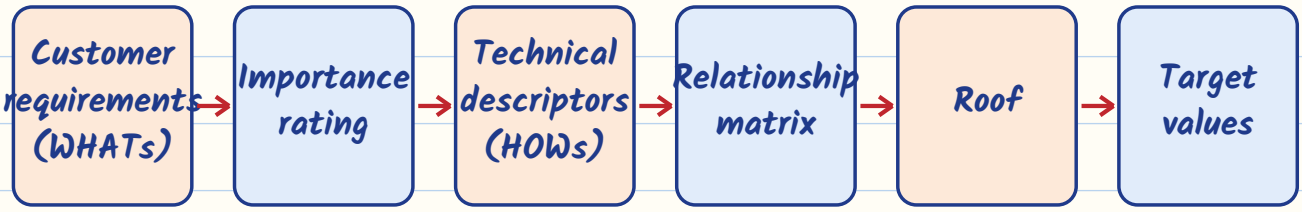
2. Quality Function Deployment (QFD)

QFD is a planning tool that translates the **voice of the customer (VOC)** into technical requirements for each stage of product development and production. Developed in Japan (Yoji Akao, 1966; Mitsubishi Kobe shipyard 1972; used by Toyota).

- Also called the "customer driven engineering" - answers **WHAT** the customer wants and **HOW** the company will provide it.
- Team approach: marketing, design, production, quality, purchase.

House of Quality (HOQ)

- The main matrix of QFD, shaped like a house.



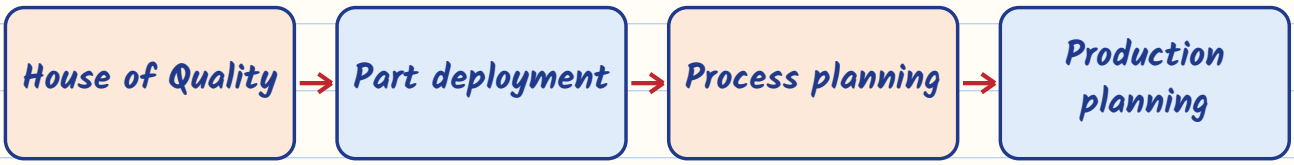
Parts of the House of Quality

1. **Left wall:** customer requirements (WHATs) from market research, with importance ratings.
2. **Ceiling:** technical requirements / design characteristics (HOWs) - measurable engineering responses.
3. **Main body (relationship matrix):** strength of relation between each WHAT and HOW: strong (9, double circle), medium (3), weak (1).
4. **Roof (correlation matrix):** relation among the HOWs - positive (+) or negative (-); to find trade-offs.
5. **Right wall (planning matrix):** customer perception, competitor comparison, sales point, improvement ratio.
6. **Basement (bottom):** technical priorities, competitor technical benchmarking and **target values.**

- Example: Customer wants "easy to open door of car" -> HOW: door

closing effort, seal compression, hinge force.

QFD process (the 4 phases)



- Phase 1: customer needs -> design requirements. Phase 2: design requirements -> part characteristics. Phase 3: part characteristics -> process operations. Phase 4: process -> production/quality control plans.
- Steps: list customer requirements -> rate importance -> list technical descriptors -> relationship matrix -> competitor evaluation -> correlation -> set targets -> select items to deploy.

Benefits of QFD

- Customer focused; reduces design changes and development time (up to 30-50%); lower cost; fewer complaints.
- Improves communication and teamwork; documents the decisions; prioritises resources; competitive benchmarking.
- Quality built in at design stage - **preventive approach.**

3. Taguchi Quality Loss Function

- **Genichi Taguchi** (Japan): Quality loss is the loss imparted to society after the product is shipped. Even if the product is **within specification limits,** any deviation from the **target (nominal) value** causes loss to the customer.
- Traditional view: loss is zero within the tolerance (LSL to USL) and then suddenly jumps (goal-post). Taguchi: **loss increases as a parabola** as the quantity departs from target.

Quadratic loss function

L(y) = k (y - T)^2

- y = actual value of the quality characteristic, T = target value, k = loss coefficient = **A / d^2** (A = cost of failure/scrap at the tolerance limit, d = tolerance half-width).
- For a population: **Average loss = k (sigma^2 + (mean - T)^2)** - loss depends on both variation and off-centring.

Types

- **Nominal-the-best:** L = k (y - T)^2 (e.g., dimension of a shaft).
- **Smaller-the-better:** L = k y^2 (e.g., wear, impurity, noise).
- **Larger-the-better:** L = k (1/y^2) (e.g., strength, mileage).

Example

- A shaft target is 20 mm, tolerance +-0.1 mm. Scrap cost at limits = Rs 50. Then k = 50 / (0.1)^2 = Rs 5000 per mm^2. For a shaft measuring 20.05 mm: L = 5000 x (0.05)^2 = **Rs 12.5.**

Importance / lessons

- Aim for the **target,** not merely inside the limits; reduce variability continuously.
- Basis for robust design, design of experiments (orthogonal arrays), signal-to-noise ratio (S/N).
- Shows quality loss as a system-wide cost - ties to the cost of quality.

Key idea: the closer to the target, the lower the loss - Quality = Uniformity around target.

4. Total Productive Maintenance (TPM)

TPM is a maintenance programme involving **all employees** that aims at maximising equipment effectiveness through a total system of preventive maintenance over the life of the equipment. (Seiichi

Nakajima, Japan, 1971; Nippondenso)

Concept

- Operator = "I operate, I maintain" **(autonomous maintenance);** maintenance staff do planned/skilled work.
- Goals: **zero breakdowns, zero defects, zero accidents.**
- TPM = TQM for equipment. Total = total participation, total effectiveness, total life cycle.

Overall Equipment Effectiveness (OEE)

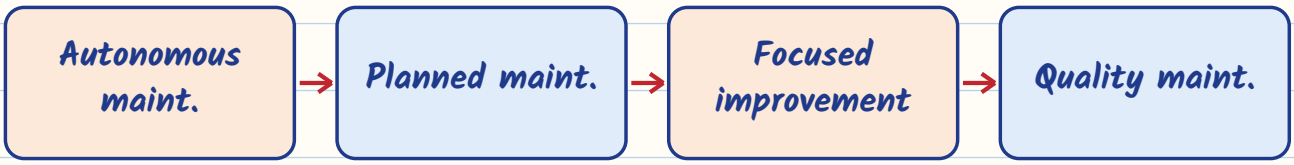
OEE = Availability x Performance rate x Quality rate

- Availability = operating time / planned time; Performance = actual speed / ideal speed; Quality = good pieces / total pieces. World class OEE = 85%.

Six big losses

1. Breakdown, 2. Setup and adjustment, 3. Idling and minor stoppages, 4. Reduced speed, 5. Process defects, 6. Start-up (reduced yield).

Eight pillars of TPM



- Also: Training and education, Early equipment management, Office TPM, Safety Health and Environment.
- Foundation is **5S.**

Improvement needs / steps in TPM implementation

- Management decision and announcement -> education -> create the organisation -> basic policy and goals -> master plan -> kick-off -> build the pillars -> **full implementation and PM award.**
- Needs: top management commitment, training, 5S, planned schedule,

spare-parts management, record of equipment history, small group activities, measure OEE.

Benefits

- Increased productivity and OEE; fewer breakdowns; lower cost; improved safety; better quality; motivated operators.

5. Failure Mode and Effects Analysis (FMEA)

FMEA is a systematic, proactive tool to identify **potential failure modes** of a product/process, their causes and effects, and to prioritise the actions to remove or reduce the risk **before** the failure occurs.

- Types: **Design FMEA (DFMEA), Process FMEA (PFMEA);** also system FMEA, service FMEA. Used in automotive (QS 9000 / IATF), aerospace, medical devices.

Stages of FMEA

1. **Form the team** and define scope; collect drawings, process flow chart.
2. Identify items/functions and list the **potential failure modes.**
3. Describe the **effects** of each failure (on customer, system).
4. Identify the **causes** of each failure mode.
5. List the **current controls** (prevention and detection).
6. Rate **Severity (S), Occurrence (O)** and **Detection (D)** on a 1-10 scale.
7. Calculate **RPN = S x O x D** (Risk Priority Number, max 1000).
8. Recommend **corrective actions** for high RPN; assign responsibility and date.
9. Implement actions, **re-evaluate** the RPN; update the FMEA (a living document).

Item	Failure mode	Effect	Cause	S	O	D	RPN

Brake pad	Wears early	Poor braking	Soft material	9	4	3	108

Rating

- Severity: 1 = no effect, 10 = hazardous without warning. Occurrence: 1 = remote, 10 = very high. Detection: 1 = almost certain to detect, 10 = cannot detect.

Benefits

- Improves reliability and safety; reduces warranty cost and late design changes; documents the knowledge; team based and prevention oriented.

Limitations

- Time consuming; subjective ratings; same RPN can come from different S, O, D combinations.

6. The Seven Tools of Quality (Old / basic QC tools - Ishikawa)

1. **Check sheet** - simple form for collecting data in real time at the place of work (tally marks). Types: defect location, defect cause, frequency check sheet.
2. **Histogram** - bar graph of frequency distribution; shows centre, spread and shape of the data (normal, skewed, bimodal).
3. **Pareto diagram** - bars in descending order with cumulative line; **80:20 rule** - vital few vs trivial many; select the problem to attack first.
4. **Cause and effect diagram (fishbone/Ishikawa)** - brainstorm causes under 6 Ms: Man, Machine, Material, Method, Measurement, Mother nature (environment); effect = head of fish. Use 5-Why for the root cause.
5. **Scatter diagram** - plot two variables to see the relationship

(positive, negative, none); correlation coefficient r.

6. **Control chart** - run chart with the centre line and UCL, LCL (+-3 sigma); separate **common cause** from **special cause** variation. Types: X-bar and R (variable), p, np, c, u (attribute).
7. **Stratification / Flow chart / Run chart** - (depending on the book: stratification, process flow chart) - separating data into groups to find the pattern.



When to use what

Tool	Answers the question
Check sheet	What is happening and how often?
Pareto	Which problem is the biggest?
Fishbone	Why is it happening?
Histogram	How is the data spread?
Scatter	Are two factors related?
Control chart	Is the process stable?

7. Process Capability

Process capability is the ability of a process to produce output **within the specification limits.** Natural process spread = **6 sigma.** Process must be in statistical control first.

- **Cp = (USL - LSL) / 6 sigma** - capability ignoring the centring (potential).
- **Cpk = min [(USL - mean) / 3 sigma , (mean - LSL) / 3 sigma]** - actual capability considering the mean shift.

- Interpretation: Cp < 1 - process not capable; Cp = 1 - just capable (0.27% outside, 2700 ppm); Cp = 1.33 - **capable** (usual minimum); Cp >= 1.67 - very capable; Cp = 2 -> Six Sigma.
- Cpk <= Cp; if Cpk = Cp the process is centred.

Example

- Specification 50 +- 3 mm: USL = 53, LSL = 47. Sigma = 0.8 and mean = 50.4. Cp = 6 / (6 x 0.8) = **1.25.** Cpk = min (2.6 / 2.4 , 3.4 / 2.4) = min (1.08, 1.42) = **1.08** - needs improvement (re-centre and reduce variation).

Steps in a capability study

- Select the characteristic -> collect 25+ subgroups -> check process is in control -> calculate mean and sigma (R-bar/d2) -> compute Cp, Cpk -> take action.

Ways to improve

- Centre the process; reduce variation (machine, material, method); widen tolerance (if permissible); 100% inspection as a last resort.

8. Concept of Six Sigma

Six Sigma is a data driven, customer focused methodology to reduce variation and defects to **3.4 defects per million opportunities (DPMO).** Introduced by Bill Smith at **Motorola (1986-87);** popularised by GE (Jack Welch).

- Sigma level: 1 sigma = 691,462 DPMO; 2 = 308,537; 3 = 66,807; 4 = 6,210; 5 = 233; **6 = 3.4** (with the 1.5 sigma shift allowed).
- DPMO = defects / (units x opportunities per unit) x 1,000,000.

DMAIC (for existing processes)

Define → **Measure** → **Analyse** → **Improve** → **Control**

- **Define** project, customer CTQ; **Measure** current performance; **Analyse** root causes; **Improve** by solution/pilot; **Control** to sustain (control plan, SPC).
- **DMADV / DFSS** for new design: Define, Measure, Analyse, Design, Verify.

Roles ("belts")

- Champion (executive sponsor), Master Black Belt, **Black Belt** (full time project leader), Green Belt (part time), Yellow Belt (team member).

Features

- Top management driven, financial benefit by project, statistical tools, training, focus on CTQ.
- Benefits: lower defects, cost saving, improved satisfaction, culture of discipline.
- Lean Six Sigma = Lean (waste removal) + Six Sigma (variation reduction).

9. New Seven Management Tools

(Used for planning and idea organising - mainly non-numerical data; developed by JUSE 1979.)

1. **Affinity diagram (KJ method)** - group large number of ideas/language data into natural categories.
2. **Interrelationship digraph** - shows the cause and effect relationship among the ideas; arrows show driver and outcome.
3. **Tree diagram** - breaks down a broad goal into progressively detailed levels of tasks (goal -> means).
4. **Matrix diagram** - shows relation between two or more groups of items (L, T, Y, X, C shapes); symbols for strength.
5. **Prioritisation matrices** - rank options by weighted criteria.

6. **Process Decision Program Chart (PDPC)** - map possible problems and countermeasures in a plan.
7. **Activity network diagram (arrow diagram / PERT-CPM)** - sequence and timing of activities, critical path.



Old tools	New tools
Numerical data	Verbal / idea data
Problem solving, control	Planning, strategy
Shop floor	Management

10. Case Studies

Case 1 - Xerox benchmarking

- Compared warehousing with L.L.Bean (generic benchmark): reduced picking time and cost. Result: significant savings and quality leadership.

Case 2 - QFD at Toyota Auto Body

- Rust problems on the car body found from VOC; QFD cut start-up and warranty cost by about 60% over 4 years.

Case 3 - TPM at a manufacturing company

- Problem: frequent breakdown of CNC machines, OEE 55%. Action: autonomous maintenance, 5S, planned PM, operator training. Result: OEE 78%, breakdowns down by 70%.

Case 4 - FMEA in automotive

- Ford (Pinto lesson): failure analysis prior to launch would have found fuel tank risk; now PFMEA is mandatory in supplier systems.

Case 5 - Six Sigma at GE / Motorola

- Savings of billions of dollars; Motorola moved from 3 sigma to near 6 sigma levels.

Important Exam Questions (Unit IV)

Short answer

- Define benchmarking. State the reasons for benchmarking.
- What is the House of Quality? Name its parts.
- Write the Taguchi loss function equation.
- What is OEE? What is RPN?
- Define Cp and Cpk. State the 7 new management tools.

Long answer

1. Explain the benchmarking process with a suitable example.
2. Explain QFD and the house of quality with a neat sketch. State the benefits.
3. Explain Taguchi's loss function and solve a numerical problem.
4. Explain TPM - concept, pillars, six big losses and OEE.
5. Explain the stages of FMEA with an example table.
6. Explain the seven QC tools with sketches.
7. Explain process capability with a numerical example. Discuss Six Sigma and DMAIC.